Building Fire-Resilience Credit Model

Poisson Initiation with an 8-State Bayesian Containment Chain

MKM Research Labs

Version 0.20 — 06-June-2026

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Contents

1 Executive Summary

The Building Fire-Resilience Credit Model (`MKM-FIRE-001`) prices the fire resilience of a commercial asset so that fire loss aggregates into the same resilience-credit currency as the existing 10,000-storm Monte Carlo engine. The credit is consumed by the Commercial PRS Pricer as an independent *full-conflagration* leg, combined with the flood and wind legs through a root-sum-of-squares all-in coupon. The model is built from four governance-separated component models:

- **Model A — Initiation.** A Poisson ignition frequency λ_i per asset, composed from a base annual rate by commercial type and multiplicative modifiers for occupancy, process load, condition, protection and fire history.
- **Model B — Intensity Track.** A deterministic 15-minute latent intensity I_t whose growth is damped by passive (structural) defence and cut once active suppression bites.
- **Model C — Response Effectiveness.** A deterministic mapping from the asset's CDM resilience levels to the dynamic quantities Models B and D consume (detection time, suppression-bite time, growth rate, critical threshold, controllability, and the fire-service-reach gate).
- **Model D — Containment, Loss & Resilience Credit.** An 8-state discrete-time Bayesian chain marched in 15-minute steps to an absorbing terminal state, aggregated into the credit currency.

The central concept is the *point of no return* (PNR): the first step at which latent intensity I_t exceeds the critical threshold I_{crit} *before* suppression has bitten — or, equivalently, the first step of any fire in a building where suppression is structurally unreachable. Once PNR latches, the containment chain switches to an absorbing matrix whose Contained column is zero: the fire can no longer be contained and absorbs into {partial loss, total loss}. The PNR frequency, $n_{\text{PNR}}/n_{\text{sim}}$, multiplied by a 100% loss-given-event, is the bps the PRS pricer charges for the fire leg.

Status: Development (Tier 2, RAG Amber). All four component models and the end-to-end orchestrator are delivered with unit tests, including the fire-service-reach interaction. Every numeric parameter is a first-pass engineering-judgement seed (`config/fire_matrices.json`, `version "seed-0"`); the results in this document are therefore *directional, not calibrated.*

Key risk: The model uses placeholder calibration throughout. It is not approved for production until historical calibration completes and the Model Risk Committee (MRC) issues sign-off (remediation items FIRE-R1–R4).

2 Model Purpose and Scope

2.1 Purpose

To assign each commercial asset a fire-resilience credit — expressed as an annual unconditional loss frequency, a conditional containment rate, and a full-conflagration (PNR) frequency — that can be priced pari-passu with the flood and wind hazards already in the portfolio engine. The credit rewards genuine, evidenced fire resilience (detection, suppression, compartmentation, tested procedures) and penalises the configurations that historically drive total losses, most notably tall buildings without adequate internal suppression.

2.2 Scope

The model covers commercial assets described by the v2 commercial-asset and resilience sections of the Common Data Model (CDM). It reads only the derived `AssetFireFeatures` bundle — never the raw CDM record — so that the feature contract is explicit, testable and stable against CDM schema churn. The in-scope drivers are: commercial type, occupancy status, business-rates category, property condition, the nine resilience-level fields (automatic detection, suppression systems, emergency procedures, structural fire resistance, compartments, fire stopping, external materials, access route, business continuity), fire-damage history, and number of storeys. The model produces a per-asset resilience credit and a portfolio roll-up; both are reproducible from a single random seed.

2.3 Out of Scope

- Physical (CFD) fire-spread and thermal-plume modelling.
- Smoke / life-safety egress and casualty modelling.
- Wildland–urban interface and external conflagration spread.
- Business-interruption quantum (only the resilience credit is produced; the loss-given-event is fixed at 100% for the conflagration leg).
- Explicit floor-level geometry — building height enters only through `NumberOfStoreys`.

3 Mathematical Framework

3.1 Initiation (Model A)

The per-asset annual ignition rate is the base rate for the commercial type scaled by five independent multipliers:

$$\lambda_i = \lambda_{0,c}\, m_{\text{occ}}\, m_{\text{proc}}\, m_{\text{cond}}\, m_{\text{protection}}\, m_{\text{history}},$$

where $\lambda_{0,c}$ is the base annual frequency by commercial type c; m_{occ} from occupancy status; m_{proc} from the process load implied by type (raised, never lowered, by a business-rates override such as *Restaurant*); m_{cond} from property condition; $m_{\text{protection}}$ from the mean resilience level across the protection fields; and m_{history} from the severity and recency of any prior fire. Unknown CDM options fall back to a neutral multiplier of 1.0, so a partial record still yields a usable λ_i.

Over a run horizon of h years the effective rate is $\lambda_{\text{eff}} = \lambda_i\, h$. For each of n_{sim} draws a Poisson count $N \sim \text{Poisson}(\lambda_{\text{eff}})$ is sampled, and a fire instantiates for that draw when $N \geq 1$ — the direct analogue of storm-event generation, where only a subset of draws produce an event. Each instantiated fire is tagged with an `InitiationClass` drawn from the per-type categorical prior (Appendix B).

3.2 Response Effectiveness (Model C)

Each resilience level is mapped to an index $k \in \{0, \ldots, 4\}$ in the CDM `RESILIENCE_LEVELS` vocabulary and to a fraction $f = k/4 \in [0, 1]$. A field-group effectiveness is the mean over its fields of the field's configured range linearly interpolated by f:

$$e_{\text{group}} = \frac{1}{|G|} \sum_{j \in G} \left[\, \ell_j + f_j \,(u_j - \ell_j) \,\right],$$

giving the aggregate passive effectiveness e_{passive} (structural resistance, compartments, fire stopping, external materials) and response effectiveness e_{response} (access route, procedures, continuity).

Detection time is $\tau_{\text{det}} = \tau_{\text{det},0}\, \delta_\ell$ where δ_ℓ is the detection multiplier for the asset's detection level. Suppression-bite time applies a gentle per-storey height penalty $p \in [p_{\min}, p_{\max}]$ interpolated over storeys, but is governed first by the **fire-service-reach gate**: external attack reaches only up to R storeys, and above R effective suppression requires internal sprinklers, i.e. a suppression level index $\geq \theta_{\text{int}}$. Writing s for the storey count and k_{supp} for the suppression-level index,

$$\text{reachable} \;=\; \big(k_{\text{supp}} \geq \theta_{\text{int}}\big) \;\vee\; \big(s \leq R\big),$$

$$\tau_{\text{supp}} = \begin{cases} \tau_{\text{supp},0}\; p & \text{if reachable,} \\ \tau_\infty \;(\gg \text{step budget}) & \text{otherwise.} \end{cases}$$

Compartmentation quality $q = \frac{1}{2}\big(k_{\text{comp}} + k_{\text{stop}}\big)$ selects the well- vs. poorly-compartmented growth rate; the suppression level selects both the post-bite growth multiplier μ_{supp} and the critical intensity I_{crit} (strong vs. weak). Finally the controllability score is the weighted blend

$$C = w_p\, e_{\text{passive}} + w_r\, e_{\text{response}} + w_s\, f_{\text{supp}},$$

(renormalised by $w_p + w_r + w_s$), which selects the pre-PNR transition matrix.

3.3 Intensity Track (Model B)

The latent intensity starts at $I_0 = 0$ and grows each 15-minute step by

$$g_t = g_{\text{base}}\,(1 - w_{\text{damp}}\, e_{\text{passive}}), \qquad g_t \leftarrow g_t \cdot \mu_{\text{supp}} \;\text{ once suppression is active,}$$

$$I_{t+1} = I_t + g_t, \qquad g_t \geq 0,$$

where g_{base} is the compartmentation-selected growth rate and suppression becomes active at the step $\lceil \tau_{\text{supp}} \rceil$. The track is fully deterministic; all stochasticity lives in the chain march (Model D).

3.4 Point of No Return

PNR latches at the first step t for which suppression is not yet active and

$$I_t > I_{\text{crit}} \quad \vee \quad \neg\,\text{reachable},$$

and the latch is sticky (once set, it stays set). The second disjunct is the operational heart of the model: a building taller than the fire-service reach with no adequate internal suppression is uncontrollable *from ignition*, so PNR latches at step 1. This is what turns building height into a live driver of total loss.

3.5 Containment Chain (Model D)

Each fire is a discrete-time chain over the eight states

$$S_0 \text{ Ignition, } S_1 \text{ Growth, } S_2 \text{ Detected, } S_3 \text{ Internal response, } S_4 \text{ External response,}$$

$$S_5 \text{ Contained, } S_6 \text{ Partial loss, } S_7 \text{ Total loss,}$$

with $\{S_5, S_6, S_7\}$ absorbing. At each step the intensity track advances and the active 8×8 row-stochastic matrix is selected: while PNR is unlatched, a hard switch on C against the threshold picks the controllable matrix ($C \geq$ threshold) or the weak matrix; once PNR latches, every transient state uses the absorbing point-of-no-return matrix, whose S_5 (Contained) column is identically zero. The next state is sampled from the current state's row (with defensive renormalisation) until an absorbing state is reached or the step budget is exhausted. The three matrices are given in Appendix D.

3.6 Resilience Credit

Aggregating over the n_{sim} draws (of which n_{fires} ignite), with terminal counts $n_{\text{contained}}$, n_{partial}, n_{total} and n_{PNR} fires that reached the point of no return, the credit currency is

$$\text{loss frequency} = \frac{n_{\text{partial}} + n_{\text{total}}}{n_{\text{sim}}}, \qquad \text{containment rate} = \frac{n_{\text{contained}}}{n_{\text{fires}}},$$

$$\text{fire spread (bps)} = \frac{n_{\text{PNR}}}{n_{\text{sim}}} \times 1.0 \times 10{,}000,$$

the last being the full-conflagration leg the Commercial PRS Pricer consumes (frequency $\times$ 100% loss-given-event). It is combined with the flood/wind legs into an all-in coupon by root-sum-of-squares, all-in $= \sqrt{\sum_{\text{perils}} \text{leg}^2}$, treating the perils as independent.

4 Input Parameters and Calibration

All numeric seeds live in `config/fire_matrices.json` (`version "seed-0"`) and are loaded by `config/fire.py`; no number is embedded in the model code. The resilience-level vocabulary is imported from `port.cdm.asset.resi` so the level-to-index-to-fraction mapping stays single-source with the rest of the CDM. The full parameter set — initiation base rates and the five modifier tables, initiation-class priors, the progression dynamics (growth, critical intensity, detection/suppression multipliers, passive/response ranges, height penalty, fire-service reach, timing, controllability weights) and the three transition matrices — is reproduced verbatim in the Appendices.

Calibration status. Every value is a first-pass engineering-judgement seed chosen for plausible *direction* and *ordering*, not magnitude: e.g. hotels and mixed-use ignite more often than offices; worse condition and worse protection raise frequency; better resilience levels shorten detection and suppression times and raise the critical intensity. Calibration against historical commercial fire-loss data (FIRE-R1) and an independent benchmarking exercise (FIRE-R2) are open remediation items; until they close, the outputs must be read as relative resilience signals, not absolute loss probabilities.

5 Implementation Details

5.1 Module Layout

- `config/fire.py` — parameter schema (enums, dataclasses, loader).
- `config/fire_matrices.json` — seed numeric parameters.
- `src/models/fire/data_structures.py` — runtime types (`AssetFireFeatures`, `ResponseProfile`, `IntensityTrack`, `ContainmentOutcome`, `AssetFireResult`).
- `src/models/fire/initiation.py` — Model A (Poisson initiation).
- `src/models/fire/response_effectiveness.py` — Model C.
- `src/models/fire/intensity_track.py` — Model B.
- `src/models/fire/containment.py` — Model D and the end-to-end orchestrator (`simulate_asset_fire`, `simulate_portfolio_fire`).
- `src/routes/commercial/hazard.py` — read-time join that attaches the fire leg (`fire_spread_bps`) to the asset hazard payload.

5.2 Randomness

All randomness flows through a single caller-owned `numpy.random.Generator`, so an entire portfolio run is reproducible from one seed — matching the storm and typhoon model convention. Model C is deterministic, Model B is deterministic, and only the chain march in Model D draws from the generator.

6 Validation and Backtesting

Historical calibration and backtesting are open remediation items (FIRE-R1, FIRE-R2) — to be completed before production sign-off. Current validation is by directional unit tests, which confirm the model behaves as designed: a strong asset contains far more often and loses far less than a weak one; the point-of-no-return gate is reachable for genuinely weak or tall-and- unsprinklered assets but not for strong ones; the fire-service-reach interaction flips a building from fully containable to a guaranteed conflagration exactly at the reach height when internal suppression is inadequate; and a whole-portfolio run is bit-for-bit reproducible from a single seed. The sensitivity tables in the next section are produced by running the *actual* end-to-end model.

6.1 Automated Test Results

Automated test-results table to be generated. Stage 1 + Stage 2 unit tests live under `tests/models/fire/` *(test_initiation.py, test_response_effectiveness.py, test_intensity_track.py, test_containment.py).*

7 Sensitivity Analysis

The tables below are generated by `docs/models/sensitivities/fire_resilience/` running the real `simulate_asset_fire` against a fixed baseline (a fully-occupied 3-storey *Office* in *Fair* condition with every resilience field at *Meets minimum*), varying one input at a time over 20,000 Monte Carlo draws per cell.

Two findings dominate. First, on the single-variable sweeps the fire-spread (PNR) leg is *zero*: the 3-storey baseline is always within the fire-service reach, so suppression is reachable and no fire runs to conflagration; the resilience levels move only the containment rate and the ordinary loss frequency, and commercial type / condition move loss frequency in the expected order (hotels, mixed-use and poor condition are worst). Second, the height $\times$ suppression grid exposes the headline non-linearity: a building *above* the 4-storey reach with only *Partial* internal suppression jumps to $\approx$ 203 bps of fire spread, while the same building at *Meets minimum* or better, or at or below the reach height, stays at zero. This cliff — not a smooth gradient — is the economically important behaviour the credit is designed to price.

Table 1: Fire-resilience credit by suppression-systems level (3-storey office baseline)

	Containment (%)	Loss freq (%)	Fire spread (bps)
Suppression: Not assessed	88.4	0.2500	0.0000
Suppression: Partial	88.7	0.2300	0.0000
Suppression: Meets minimum	88.7	0.2300	0.0000
Suppression: Enhanced	88.7	0.2300	0.0000
Suppression: Verified	88.3	0.2300	0.0000

Table 2: Fire-resilience credit by building height (reach = 4 storeys; 'Meets minimum' suppression)

	Containment (%)	Loss freq (%)	Fire spread (bps)
1 storeys	88.7	0.2300	0.0000
3 storeys	88.7	0.2300	0.0000
4 storeys	88.7	0.2300	0.0000
5 storeys	88.7	0.2300	0.0000
8 storeys	88.7	0.2300	0.0000
12 storeys	88.7	0.2300	0.0000
20 storeys	88.7	0.2300	0.0000

Table 3: Fire-resilience credit by compartmentation level

	Containment (%)	Loss freq (%)	Fire spread (bps)
Compartments: Not assessed	88.7	0.2300	0.0000
Compartments: Partial	88.7	0.2300	0.0000
Compartments: Meets minimum	88.7	0.2300	0.0000
Compartments: Enhanced	88.7	0.2300	0.0000
Compartments: Verified	88.7	0.2300	0.0000

Table 4: Fire-resilience credit by property condition

	Containment (%)	Loss freq (%)	Fire spread (bps)
Excellent	88.5	0.1900	0.0000
Good	87.9	0.2300	0.0000
Fair	88.7	0.2300	0.0000
Poor	87.9	0.2850	0.0000
Very poor	88.0	0.3300	0.0000

Table 5: Fire-resilience credit by commercial type

	Containment (%)	Loss freq (%)	Fire spread (bps)
Office	88.7	0.2300	0.0000
Retail	87.6	0.4150	0.0000
Hotel	87.7	0.5300	0.0000
Leisure	87.8	0.4550	0.0000
Healthcare	87.6	0.3500	0.0000
MultiFamily	87.9	0.3900	0.0000
MixedUse	87.4	0.5700	0.0000
Other	87.4	0.5700	0.0000

Table 6: Fire spread (bps) by height × suppression level — the fire-service-reach interaction

	Partial	Meets minimum	Enhanced	Verified
3 storeys	0.0000	0.0000	0.0000	0.0000
5 storeys	203.0	0.0000	0.0000	0.0000
8 storeys	203.0	0.0000	0.0000	0.0000
20 storeys	203.0	0.0000	0.0000	0.0000

8 Model Limitations and Known Weaknesses

- **No historical calibration (FIRE-L1).** Every parameter is a first-pass placeholder; outputs are relative signals, not absolute loss probabilities.
- **Height proxy (FIRE-L2).** Building height is proxied by `NumberOfStoreys`; the commercial schema carries no explicit floor-level or floor-area metric, and the fire-service reach is a single global threshold rather than a brigade-specific capability.

- **Deterministic intensity track (FIRE-L3).** All stochasticity lives in the chain march; the intensity growth and PNR timing are deterministic given the asset profile.
- **Initiation class is cosmetic.** The initiation-class draw seeds an entry-point label but does not yet change the chain start state or dynamics.
- **Fixed loss-given-event.** The conflagration leg assumes a 100% loss given a point-of-no-return event; partial-loss quantum is not modelled.
- **Independence assumption.** The root-sum-of-squares all-in coupon treats fire, flood and wind as independent perils.

9 Governance Validation Questions

The following nine questions are mandated by Chapter 5 of the *Model Risk Governance for Vendors* handbook and must be explicitly addressed for every model in the inventory. Response status is tracked in the Model Governance Dashboard (MRC portal).

9.1 Q1 — Model purpose and use

"What is the model used for, and is that use consistent with its design?"

The model produces a fire-resilience credit (loss frequency, containment rate and a full-conflagration bps leg) consumed by the Commercial PRS Pricer as an independent peril. Use is consistent with design; it is *not* a life-safety or absolute-loss model and must not be used as one.

9.2 Q2 — Data quality and lineage

"Are the model inputs of sufficient quality, and is their lineage documented?"

Inputs are the CDM-derived `AssetFireFeatures`; lineage runs from the v2 CDM resilience fields through Model C. Missing options degrade gracefully to neutral multipliers. Data *quality* for calibration (historical fire-loss records) is not yet established — open under FIRE-R1.

9.3 Q3 — Conceptual soundness

"Is the methodology theoretically sound and appropriate?"

The four-model decomposition (Poisson initiation, deterministic intensity track, resilience→dynamics mapping, absorbing Bayesian chain) is a standard and defensible structure. The point-of-no-return gate is the one bespoke mechanism and is documented in Section 3.4.

9.4 Q4 — Calibration and parameters

"How were parameters estimated, and are they appropriate?"

All parameters are engineering-judgement seeds (`version "seed-0"`) chosen for direction and ordering, not magnitude. Statistical calibration is an open remediation item (FIRE-R1, FIRE-R2). This is the dominant model limitation.

9.5 Q5 — Implementation verification

"Has the implementation been verified against the specification?"

Yes, by unit tests covering each component model, the orchestrator and the fire-service-reach interaction, plus a reproducibility test. The sensitivity harness exercises the full end-to-end path.

9.6 Q6 — Sensitivity and stability

"How sensitive are outputs to inputs and assumptions?"

See Section 7. Outputs respond in the expected order to type, condition and resilience levels, and exhibit the designed height×suppression cliff. Stability under draw count is adequate at $n_{\text{sim}} = 20{,}000$; formal stability bands are an open item.

9.7 Q7 — Limitations and weaknesses

"Are the model's limitations understood and documented?"

Yes — see Section 8 (FIRE-L1–L3 plus four further documented assumptions). The placeholder calibration and the height proxy are the most material.

9.8 Q8 — Ongoing monitoring

"How will the model be monitored in production?"

Not yet in production. A monitoring plan (drift in input distributions, backtest of realised vs. predicted fire frequency) will be defined as part of FIRE-R3 before sign-off.

9.9 Q9 — Governance and approval

"What is the approval status and who owns the model?"

Owner: David K Kelly. Status: Development, Tier 2, RAG Amber. Not approved for production; MRC sign-off is contingent on closing FIRE-R1–R4.

10 Change History

Version	Date	Description
0.01	05-June-2026	Skeleton document. Stage 1 (Model A) and Stage 2 (Models B/C/D + orchestrator) delivered with unit tests; placeholder calibration.
0.20	06-June-2026	Completed all narrative sections; added the full mathematical framework (Models A–D, PNR, credit currency), governance validation questions, the matrices appendix, and the end-to-end sensitivity analysis.

A Initiation Parameters (Model A)

Table 7: Base annual ignition frequency $\lambda_{0,c}$ by commercial type.

Commercial type	$\lambda_{0,c}$
Office	0.020
Retail	0.030
Hotel	0.035
Leisure	0.030
Healthcare	0.025
MultiFamily	0.030
MixedUse	0.040
Other	0.040

Table 8: Frequency modifiers. Process load is by type with a business-rates override (*Restaurant* = 1.25) that only ever raises the value.

Modifier	Option	Value
m_{occ} (occupancy)	Fully occupied	1.00
	Partially vacant	0.90
	Vacant	0.85
m_{proc} (process)	Office	1.00
	Retail	1.10
	Hotel	1.25
	Leisure	1.25
	Healthcare	1.10
	MultiFamily	1.05
	MixedUse	1.15
	Other	1.15
m_{cond} (condition)	Excellent	0.85
	Good	0.95
	Fair	1.05
	Poor	1.20
	Very poor	1.40
$m_{\text{protection}}$ (level)	Not assessed	1.10
	Partial	1.00
	Meets minimum	0.95
	Enhanced	0.90
	Verified	0.85
m_{history}	No prior	1.00
	Old minor	1.05
	Recent minor	1.10
	Recent moderate/severe	1.40

A fire counts as *recent* if it occurred within `recent_years` = 5 years.

B Initiation-Class Priors

Table 9: Categorical priors over initiation class by commercial type (IE = InternalElectrical, KO = KitchenOccupant, PT = PlantTechnical, PR = ProcessRelated, BS = BasementService, EE = ExternalExposure, Oth = Other). Rows sum to 1.

Type	IE	KO	PT	PR	BS	EE	Oth
Office	0.35	0.10	0.15	0.05	0.15	0.10	0.10
Retail	0.30	0.30	0.10	0.05	0.10	0.10	0.05
Hotel	0.28	0.25	0.15	0.05	0.12	0.10	0.05
Leisure	0.30	0.22	0.13	0.05	0.10	0.10	0.10
Healthcare	0.32	0.12	0.20	0.06	0.12	0.08	0.10
MultiFamily	0.35	0.30	0.08	0.02	0.10	0.08	0.07
MixedUse	0.28	0.12	0.15	0.12	0.10	0.08	0.15
Other	0.28	0.12	0.15	0.12	0.10	0.08	0.15

C Progression Dynamics (Models B & C)

Table 10: Growth, critical intensity, and per-level detection / suppression multipliers.

Block	Key	Value
Growth per step	well_compartmented	1.5
	poorly_compartmented	4.0
I_{crit}	strong_suppression	75
	weak_suppression	12
Detection mult. δ_ℓ	Not assessed	1.00
	Partial	0.90
	Meets minimum	0.80
	Enhanced	0.72
	Verified	0.65
Suppression growth mult. μ_{supp}	Not assessed	1.00
	Partial	0.85
	Meets minimum	0.75
	Enhanced	0.62
	Verified	0.50

Table 11: Passive and response effectiveness ranges $[\ell, u]$, interpolated by the field's resilience-level fraction.

Group	Field	ℓ	u
Passive	StructuralFireResistanceAdequate	0.60	0.95
	CompartmentsProvided	0.55	0.90
	FireStoppingAtPenetrations	0.70	0.95
	ExternalMaterialsFireResistant	0.65	0.95
Response	AccessRouteResilient	0.70	0.95
	EmergencyProceduresTested	0.80	0.97
	BusinessContinuityPlanInPlace	0.85	0.98

Table 12: Height penalty, fire-service reach, timing, intensity dynamics and controllability.

Block	Key	Value
Height penalty	range	[1.05, 1.60]
	max_storeys	20
Fire-service reach	reach_storeys R	4
	internal_suppression_threshold θ_{int}	2
	no_reach_bite_steps τ_∞	999.0
Timing	detection_base_steps $\tau_{\text{det},0}$	2.0
	suppression_bite_base_steps $\tau_{\text{supp},0}$	3.0
Intensity dynamics	passive_damp_weight w_{damp}	0.60
	compartmentation_well_threshold	2.0
	strong_suppression_threshold	2.0
Controllability	weight (passive) w_p	0.35
	weight (response) w_r	0.35
	weight (suppression) w_s	0.30
	switch_threshold	0.55

D Containment Transition Matrices (Model D)

Each is an 8×8 row-stochastic matrix over $\{S_0, \ldots, S_7\}$; the step length is 15 minutes. The point-of-no-return matrix has a zero S_5 (Contained) column — once latched, containment is unreachable.

Table 13: `AA_controllable` — pre-PNR, controllable asset ($C \geq 0.55$).

From\To	S_0	S_1	S_2	S_3	S_4	S_5	S_6	S_7
S_0	0.05	0.25	0.70	0	0	0	0	0
S_1	0	0.10	0.85	0	0	0	0.05	0
S_2	0	0	0	0.90	0.08	0	0.02	0
S_3	0	0	0	0	0.10	0.85	0.05	0
S_4	0	0	0	0	0	0.75	0.20	0.05
S_5	0	0	0	0	0	1.00	0	0
S_6	0	0	0	0	0	0	0.85	0.15
S_7	0	0	0	0	0	0	0	1.00

Table 14: `weak_controllable` — pre-PNR, weak asset ($C < 0.55$).

From\To	S_0	S_1	S_2	S_3	S_4	S_5	S_6	S_7
S_0	0.05	0.70	0.25	0	0	0	0	0
S_1	0	0.10	0.45	0	0	0	0.45	0
S_2	0	0	0	0.40	0.40	0	0.20	0
S_3	0	0	0	0	0.15	0.15	0.70	0
S_4	0	0	0	0	0	0.25	0.50	0.25
S_5	0	0	0	0	0	1.00	0	0
S_6	0	0	0	0	0	0	0.50	0.50
S_7	0	0	0	0	0	0	0	1.00

Table 15: `point_of_no_return` — absorbing, post-PNR. The S_5 column is identically zero.

From\To	S_0	S_1	S_2	S_3	S_4	S_5	S_6	S_7
S_0	0	1.00	0	0	0	0	0	0
S_1	0	0	0	0	0	0	0.40	0.60
S_2	0	0	0	0	0	0	0.40	0.60
S_3	0	0	0	0	0	0	0.40	0.60
S_4	0	0	0	0	0	0	0.40	0.60
S_5	0	0	0	0	0	1.00	0	0
S_6	0	0	0	0	0	0	0.35	0.65
S_7	0	0	0	0	0	0	0	1.00